Supplementary Material: Sensor Metadata and Axis-Aware Tokenization Shape Zero-Shot Reconstruction in OPM-MEG

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This supplement presents additional methods, participant-level reconstruction results, paired comparisons, decoding results, and dataset details.

S1 Soft Homogeneous Field Correction

S1.1 Setup

Following Tierney et al. (2021), we approximate distant magnetic interference as a spatially uniform field $\mathbf{B}_{\text{ext}} \in \mathbb{R}^3$. A sensor c with unit sensitive orientation $\mathbf{u}_c$ measures the projection $m_c = \mathbf{u}_c^\top \mathbf{B}_{\text{ext}}$, so across an array of C channels the induced measurement is $\mathbf{m}_{\text{ext}} = \mathbf{Q}\mathbf{B}_{\text{ext}}$, where row c of $\mathbf{Q} \in \mathbb{R}^{C\times 3}$ is $\mathbf{u}_c^\top$.

Conventional homogeneous field correction (Tierney et al. 2021) projects onto the null space of $\mathbf{Q}^\top$:

$$\mathbf{P}_{\text{hard}} = \mathbf{I}_C - \mathbf{Q}(\mathbf{Q}^\top\mathbf{Q})^{-1}\mathbf{Q}^\top.$$

When $\mathbf{Q}$ has full column rank, $\operatorname{rank}(\mathbf{P}_{\text{hard}}) = C-3$, making the projected channel covariance rank-deficient.

The reference implementation of this correction is `mne.preprocessing.compute_proj_hfc` in MNE-Python (MNE-Python Developers n.d.), whose `order` argument selects the homogeneous field alone or a higher-order harmonic basis (Tierney et al. 2022). Seymour et al. (2022) survey the interference-suppression options available for OPM arrays. We did not use that implementation: the variant evaluated here applies partial rather than full projection and is defined next.

S1.2 Soft variant

The cited method uses full projection. Here we evaluate a soft shrinkage variant parameterized by $\alpha \in [0, 1)$:

$$\mathbf{P}_{\text{soft}}(\alpha) = \mathbf{I}_C - \alpha\,\mathbf{Q}_R\mathbf{Q}_R^\top,$$

where $\mathbf{Q}_R$ is an orthonormal basis of $\operatorname{col}(\mathbf{Q})$, of dimension $r = \operatorname{rank}(\mathbf{Q}) \le 3$. Its eigenvalues are $1-\alpha$ on those r directions and 1 on the remaining $C-r$, so for any $\alpha < 1$ the operator has full rank C. The current implementation uses a rank-aware SVD to avoid extra QR directions for rank-deficient arrays. Historical results used thin QR and have not been recomputed after this correction.

The coefficient $\alpha = 0.25$ is an analysis choice here, not a value established by Tierney et al. (2021). At this value, components lying in the uniform-field subspace are scaled by 0.75 in amplitude, a power reduction of $1-0.75^2 = 43.75\%$.

S1.3 What this does to neural signal

A neural component orthogonal to $\operatorname{col}(\mathbf{Q})$ satisfies $\mathbf{Q}_R^\top\mathbf{s} = \mathbf{0}$ and is returned unchanged. This is not a claim that neural signal is preserved. A dipolar field pattern is not in general orthogonal to the uniform-field subspace, and whatever part of it does project onto $\operatorname{col}(\mathbf{Q})$ receives the same 25% amplitude shrinkage as the interference. Soft-HFC is therefore a lossy operation on signal as well as noise, which is why the main paper holds filtering constant within every contrast rather than treating it as a neutral preprocessing step.

S2 Single-Axis Metadata Interventions

All conditions share channel selection, trials and Soft-HFC. Only the metadata handed to the tokenizer differs, so any difference between columns of Table S1 is a property of the learned embedding rather than of the recording. Table S2 gives the paired tests.

The dataset documentation states that FACE-DEC sensor coordinates were approximated from an Elekta layout for topographic display and are not measured OPM positions. The geometry rows should therefore be read as sensitivity to supplied metadata, not as evidence about forward-field accuracy.

S3 Tag by Partition Factorial

The experiment crosses MAG/GRAD tags, raw/Soft-HFC inputs and monolithic/random/axis partitions. Random group sizes match each participant's Z, X and Y counts exactly in the generating script, with seeds 42 and 101–109. Saved group assignments are unavailable for an independent metadata-level check. We average seeds within participant before paired inference, retaining ten independent participants. Tables S3–S6 report the full-array reconstruction results.

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PCC is averaged over channels and temporal windows within each trial, then over trials within participant. Each participant contributes equally to cohort estimates.

The tag effect is configuration-dependent. The raw monolithic contrast is $+0.0002$ ($[-0.0104, +0.0106]$, $p = 0.969$), while axis grouping retains its advantage under both tags. The paired change in the grouping advantage under GRAD is -0.0432 raw and -0.0326 with Soft-HFC, both negative in every participant. This interaction does not by itself identify a shared mechanism. Under Soft-HFC, the monolithic-to-random contrast accounts for 23.5% under MAG and 40.5% under GRAD of the respective axis-over-monolithic gains. These are descriptive partition components: random partitioning changes attention composition as well as query budget, so a capacity contribution cannot be isolated. Tests are exploratory and unadjusted for multiple comparisons.

S4 Decoding

The classical decoder uses scikit-learn `LogisticRegression` with L_2 penalty at the library default $C = 1.0$, fixed for every configuration and never tuned on a held-out fold, so no selection leakage is possible. The pretrained probe uses the same solver and folds. The sensor-type codes are checkpoint-specific integers, `MAG=1` and `GRAD=2` in the released tiny configuration, and any reuse of this intervention must read them from the checkpoint rather than assume them.

Table S7 gives all twelve full-array configurations and three decoders, evaluated with five-fold cross-validation within participant. Random-partition decoding uses seed 42 with group sizes matched to the axes, while reconstruction averages ten seeds. Every pretrained-probe interval includes chance. Table S8 gives the paired axis-minus-monolithic and axis-minus-random probe contrasts; every interval includes zero. These are exploratory tests without multiplicity correction, not evidence of equivalence.

The classical features are six log band powers spanning 1–96 Hz plus four temporal window means per channel. They depend on filtering but not on sensor tag or tokenizer partition. The six classical rows within each filtering regime therefore share one underlying measurement. The Soft-HFC interval excludes chance, while the unfiltered interval does not. The classical cohort mean exceeds the pretrained probe in every configuration, but this ordering alone does not establish a significant paired advantage or rule out a task-level noise floor.

S5 Model Architecture

The pretrained BrainOmni tiny configuration is evaluated with frozen weights. The tokenizer conditions on sensor position, unit orientation, and a discrete sensor type, and uses residual vector quantization (Xiao et al. 2025). In the evaluated configuration, the sequence backbone is bidirectional (`causal=False`) and uses masked-token prediction over codebook entries with a mask ratio of 0.25. Attention operates within trials, not across trials. The number of learned spatial queries is loaded from the checkpoint.

Axis partitioning is an inference-time reorganization only. The X, Y and Z groups pass separately through the same frozen tokenizer and the reconstructions are reassembled, giving three spatial-attention passes instead of one. No weights change.

S6 Dataset Details

FACE-DEC, OpenNeuro `ds005107` v1.0.0, DOI `10.18112/openneuro.ds005107.v1.0.0`, 21 participants and the 50 channels matched across participants by rounded position and orientation metadata. Triaxial OPM, Zenodo record `10.5281/zenodo.11126593`, the 10 adult participants, 84 cached trials each, 155 to 176 retained channels. SQUID control, OpenNeuro `ds000117` v1.0.4, participant `sub-01`, run `01`.

Participant	Default	GRAD tag	Zero ori.	Rev. ori.	Scrambled
P01	0.2717	0.3950	0.2600	0.2563	0.2327
P02	0.2522	0.3735	0.2413	0.2418	0.2331
P03	0.2435	0.3713	0.2299	0.2268	0.2265
P04	0.2444	0.3976	0.2310	0.2250	0.2420
P05	0.2484	0.3944	0.2386	0.2445	0.2466
P06	0.2573	0.3910	0.2461	0.2439	0.2279
P07	0.2580	0.3940	0.2501	0.2511	0.2350
P08	0.2630	0.3890	0.2518	0.2502	0.2327
P09	0.2641	0.3736	0.2549	0.2504	0.2382
P10	0.2486	0.3708	0.2352	0.2306	0.2376
P11	0.2546	0.3855	0.2384	0.2384	0.2300
P12	0.2415	0.3839	0.2270	0.2241	0.2355
P13	0.2302	0.3680	0.2134	0.2063	0.2347
P14	0.2579	0.3864	0.2431	0.2425	0.2352
P15	0.2352	0.3753	0.2159	0.2109	0.2251
P16	0.2571	0.3836	0.2462	0.2446	0.2350
P17	0.2564	0.3884	0.2467	0.2425	0.2251
P18	0.2509	0.3745	0.2327	0.2307	0.2303
P19	0.2562	0.3813	0.2363	0.2370	0.2268
P20	0.2373	0.3771	0.2163	0.2137	0.2258
P21	0.2226	0.3647	0.1990	0.2020	0.2368
Mean	0.2501	0.3819	0.2359	0.2340	0.2330

Table S1: Zero-shot reconstruction PCC per participant on FACE-DEC (ds005107), 50 shared channels, Soft-HFC $\alpha = 0.25$. Participants are indexed in the saved record order, which is not the dataset's own participant numbering. Waveforms are identical across columns; only the metadata supplied to the tokenizer differs.

Contrast vs default	Mean Δ	95% CI	Consistent	Exact p
Gradiometer tag	$+0.1318$	$[+0.1275, +0.1360]$	21/21	9.54e-07
Zero orientation	-0.0142	$[-0.0160, -0.0124]$	21/21	9.54e-07
Reversed orientation	-0.0161	$[-0.0183, -0.0139]$	21/21	9.54e-07
Scrambled geometry	-0.0171	$[-0.0225, -0.0115]$	19/21	4.58e-05

Table S2: Paired single-axis contrasts, 21 participants. Confidence intervals are percentile bootstrap over participants, 20,000 resamples, seed 0. The consistency column counts participants moving in the stated direction. p is the exact two-sided sign-flip permutation value enumerating all 2^{21} assignments.

Filter	Tag	Partition	Mean PCC	95% CI
Raw	MAG	Monolithic	0.3726	[0.3409, 0.4047]
Raw	MAG	Random	0.4019	[0.3721, 0.4316]
Raw	MAG	Axis	0.5138	[0.4846, 0.5441]
Raw	GRAD	Monolithic	0.3728	[0.3388, 0.4064]
Raw	GRAD	Random	0.4174	[0.3890, 0.4454]
Raw	GRAD	Axis	0.4862	[0.4652, 0.5066]
Soft-HFC	MAG	Monolithic	0.3456	[0.3102, 0.3807]
Soft-HFC	MAG	Random	0.3785	[0.3457, 0.4105]
Soft-HFC	MAG	Axis	0.4856	[0.4512, 0.5201]
Soft-HFC	GRAD	Monolithic	0.3471	[0.3141, 0.3796]
Soft-HFC	GRAD	Random	0.3978	[0.3702, 0.4247]
Soft-HFC	GRAD	Axis	0.4723	[0.4500, 0.4944]

Table S3: Full-array factorial reconstruction. Trial-weighted participant PCC; random scores average ten seeds within participant. All intervals use 20,000 participant bootstrap draws with seed 0. Neither seeds nor trials are treated as independent participants.

Filter	Contrast	Mean Δ	95% CI	Positive	Exact p
Raw	MAG: Axis minus random	$+0.1120$	$[+0.1048, +0.1208]$	10/10	1.95e-03
Raw	MAG: Random minus mono	$+0.0292$	$[+0.0262, +0.0322]$	10/10	1.95e-03
Raw	GRAD: Axis minus random	$+0.0688$	$[+0.0567, +0.0805]$	10/10	1.95e-03
Raw	GRAD: Random minus mono	$+0.0446$	$[+0.0390, +0.0504]$	10/10	1.95e-03
Raw	GRAD minus MAG: Monolithic	$+0.0002$	$[-0.0104, +0.0106]$	5/10	9.69e-01
Raw	GRAD minus MAG: Random	$+0.0155$	$[+0.0065, +0.0234]$	9/10	1.37e-02
Raw	GRAD minus MAG: Axis	-0.0277	$[-0.0408, -0.0150]$	1/10	5.86e-03
Raw	Grouping interaction	-0.0432	$[-0.0548, -0.0329]$	0/10	1.95e-03
Soft-HFC	MAG: Axis minus random	$+0.1071$	$[+0.1002, +0.1143]$	10/10	1.95e-03
Soft-HFC	MAG: Random minus mono	$+0.0328$	$[+0.0292, +0.0363]$	10/10	1.95e-03
Soft-HFC	GRAD: Axis minus random	$+0.0745$	$[+0.0656, +0.0833]$	10/10	1.95e-03
Soft-HFC	GRAD: Random minus mono	$+0.0507$	$[+0.0448, +0.0564]$	10/10	1.95e-03
Soft-HFC	GRAD minus MAG: Monolithic	$+0.0014$	$[-0.0108, +0.0120]$	6/10	8.24e-01
Soft-HFC	GRAD minus MAG: Random	$+0.0193$	$[+0.0083, +0.0287]$	9/10	1.17e-02
Soft-HFC	GRAD minus MAG: Axis	-0.0133	$[-0.0276, +0.0008]$	5/10	1.15e-01
Soft-HFC	Grouping interaction	-0.0326	$[-0.0431, -0.0233]$	0/10	1.95e-03

Table S4: Paired factorial contrasts. Positive counts strictly positive participant differences, including when the mean is negative. The grouping interaction is $(A - R)_{\mathrm{GRAD}} - (A - R)_{\mathrm{MAG}}$, with R averaged across seeds within participant. Exact two-sided sign-flip tests are exploratory and unadjusted for multiplicity.

Filter	Participant	MAG mono	MAG rand	MAG axis	GRAD mono	GRAD rand	GRAD axis
Raw	P01	0.3183	0.3502	0.4505	0.3285	0.3788	0.4459
Raw	P02	0.3391	0.3675	0.4694	0.3162	0.3675	0.4499
Raw	P03	0.3060	0.3429	0.4520	0.3009	0.3597	0.4386
Raw	P04	0.3066	0.3370	0.4821	0.2989	0.3571	0.4610
Raw	P05	0.4338	0.4579	0.5771	0.4601	0.4896	0.5253
Raw	P06	0.4617	0.4824	0.5969	0.4315	0.4672	0.5320
Raw	P07	0.4046	0.4287	0.5250	0.4008	0.4425	0.4847
Raw	P08	0.3740	0.4076	0.5122	0.3962	0.4374	0.5175
Raw	P09	0.3856	0.4144	0.5302	0.3905	0.4323	0.5088
Raw	P10	0.3967	0.4302	0.5430	0.4046	0.4419	0.4981
Soft-HFC	P01	0.2828	0.3206	0.4110	0.3029	0.3602	0.4313
Soft-HFC	P02	0.3008	0.3357	0.4317	0.2894	0.3471	0.4333
Soft-HFC	P03	0.2722	0.3146	0.4148	0.2767	0.3416	0.4236
Soft-HFC	P04	0.2721	0.3075	0.4358	0.2782	0.3398	0.4403
Soft-HFC	P05	0.4026	0.4312	0.5526	0.4312	0.4651	0.5181
Soft-HFC	P06	0.4408	0.4627	0.5766	0.3997	0.4438	0.5266
Soft-HFC	P07	0.3863	0.4120	0.5082	0.3766	0.4240	0.4771
Soft-HFC	P08	0.3536	0.3900	0.4941	0.3677	0.4215	0.4977
Soft-HFC	P09	0.3664	0.3991	0.5108	0.3709	0.4143	0.4931
Soft-HFC	P10	0.3786	0.4112	0.5203	0.3776	0.4202	0.4819

Table S5: Full-array factorial PCC per participant. Indices follow sorted dataset participant IDs. Random columns average the ten saved seeds; other columns use one condition per participant.

Seed	Raw MAG	Raw GRAD	Soft-HFC MAG	Soft-HFC GRAD
42	+0.1124	+0.0663	+0.1083	+0.0745
101	+0.1110	+0.0706	+0.1062	+0.0768
102	+0.1114	+0.0713	+0.1057	+0.0768
103	+0.1115	+0.0636	+0.1066	+0.0683
104	+0.1147	+0.0684	+0.1085	+0.0742
105	+0.1215	+0.0781	+0.1178	+0.0836
106	+0.1136	+0.0725	+0.1083	+0.0781
107	+0.1009	+0.0591	+0.0969	+0.0638
108	+0.1140	+0.0677	+0.1092	+0.0739
109	+0.1087	+0.0699	+0.1040	+0.0754
SD	0.0049	0.0049	0.0049	0.0051

Table S6: Seed-specific cohort axis-minus-random PCC contrasts, trial-weighted. SD is the descriptive population SD across the ten observed seed-specific means (divisor ten), not a standard error. Group sizes exactly match axis counts in the generating script. Every participant contrast is positive for every seed in all four tag/filter combinations.

Filter	Tag	Partition	Probe [95% CI]	Ctrl A	Classical [95% CI]	Cl. > Pr.
Raw	MAG	Monolithic	48.36 [43.83, 52.68]	49.12	53.54 [49.93, 56.75]	8/10
Raw	MAG	Random	49.40 [45.00, 53.26]	50.74	53.54 [49.93, 56.75]	7/10
Raw	MAG	Axis	47.53 [43.18, 51.85]	49.53	53.54 [49.93, 56.75]	7/10
Raw	GRAD	Monolithic	48.89 [41.83, 54.68]	48.25	53.54 [49.93, 56.75]	6/10
Raw	GRAD	Random	51.79 [47.35, 55.36]	49.56	53.54 [49.93, 56.75]	6/10
Raw	GRAD	Axis	49.93 [45.15, 54.58]	48.83	53.54 [49.93, 56.75]	7/10
Soft-HFC	MAG	Monolithic	49.07 [44.89, 52.79]	49.69	54.50 [51.03, 57.65]	8/10
Soft-HFC	MAG	Random	50.14 [45.10, 54.54]	50.22	54.50 [51.03, 57.65]	6/10
Soft-HFC	MAG	Axis	50.19 [43.71, 56.36]	48.49	54.50 [51.03, 57.65]	8/10
Soft-HFC	GRAD	Monolithic	50.32 [44.93, 55.46]	49.25	54.50 [51.03, 57.65]	6/10
Soft-HFC	GRAD	Random	50.88 [45.93, 55.43]	51.21	54.50 [51.03, 57.65]	7/10
Soft-HFC	GRAD	Axis	51.88 [46.99, 56.51]	48.97	54.50 [51.03, 57.65]	7/10

Table S7: Full-array factorial Braille balanced accuracy (%), with participant bootstrap intervals. Ctrl A denotes the random-initialization control. Random decoding is available only at seed 42, with group sizes exactly matched to the axes; reconstruction averages ten seeds. Every pretrained-probe interval includes chance; this does not establish equivalence.

Filter	Tag	Probe contrast	Mean Δ	95% CI	Positive	Exact p
Raw	MAG	Axis minus mono	-0.8333	$[-4.1111, +2.3333]$	4/10	6.45e-01
Raw	MAG	Axis minus random	-1.8750	$[-5.3889, +1.2083]$	5/10	3.52e-01
Raw	GRAD	Axis minus mono	$+1.0417$	$[-3.1111, +5.3194]$	5/10	6.56e-01
Raw	GRAD	Axis minus random	-1.8611	$[-5.5000, +1.6389]$	4/10	3.57e-01
Soft-HFC	MAG	Axis minus mono	$+1.1250$	$[-1.5694, +4.1528]$	5/10	5.29e-01
Soft-HFC	MAG	Axis minus random	$+0.0556$	$[-4.4444, +4.7087]$	4/10	9.90e-01
Soft-HFC	GRAD	Axis minus mono	$+1.5556$	$[-1.0833, +4.2778]$	7/10	3.09e-01
Soft-HFC	GRAD	Axis minus random	$+1.0000$	$[-1.3333, +3.2917]$	6/10	4.45e-01

Table S8: Paired pretrained-probe decoding contrasts in balanced-accuracy percentage points, ten participants. Random partitions use seed 42. Intervals use 20,000 participant bootstrap draws; exact two-sided sign-flip tests are exploratory and unadjusted for multiplicity. Positive counts strictly positive participant differences. Every interval includes zero; this does not establish equivalence.

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